

Machine Learning is a subset of artificial intelligence.

It enables systems to learn from data without being explicitly programmed.

Supervised learning uses labeled data to train models.

Unsupervised learning finds hidden patterns in unlabeled data.

Gradient descent is an optimization algorithm used to minimize loss functions.

A neural network is composed of layers of interconnected nodes called neurons.

Backpropagation is the algorithm used to train neural networks.

Overfitting occurs when a model learns noise instead of the underlying pattern.

Regularization techniques such as dropout help prevent overfitting.

Cross-validation is used to evaluate model generalization.